The EBMT Handbook

Hematopoietic Cell Transplantation and Cellular Therapies

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Editors

EBMT

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Preface

Hematopoietic cell transplantation (HCT) has become a well-established life-saving treatment procedure for many patients with hematological malignancies, inborn errors, or bone marrow failure syndromes. Starting more than 60 years ago as an "ultima ratio" option, HCT is now integrated as an essential part in many treatment concepts and protocols. The rapid evolution in this exciting field of medicine with innovative treatment concepts and the increasing number of long-term survivors require a continuous education of physicians, nurses, and healthcare providers who are involved in HCT and cellular therapies.

This thoroughly revised second edition of *the EBMT Handbook* with its new format is a major part of a broader educational strategy of the EBMT. This Handbook addresses the most recent developments and innovations in HCT and cellular therapy, presented by more than 200 authors, known as experts and well-recognized authorities in the field. In 94 chapters, all types of hematopoietic cell and bone marrow transplantation including haplo-identical and cord blood transplantation, indication for transplantation and management of complications as well as the new rapidly evolving field of cellular therapies are covered. Other important issues such as quality management and JACIE accreditation, stem cell collection, conditioning, donor selection, HLA typing, graft manipulation, ethical issues, psychological support, and quality of life are also thoroughly addressed.

The aim of this Handbook is, as the name implies, not an in-depth knowledge base like a textbook but rather a hands-on and concise source of information at bedside to present the state of the art and improve our practice skills. Major key points are summarized at the end of each chapter for a quick reference.

The EBMT Board wants to express their great gratitude to the strong effort of the Working Party chairs and all authors in planning and writing the chapters and the tremendous work of the project leader Isabel Sánchez-Ortega and

the secretarial work of the EBMT executive office in Barcelona but also to Karthik Periyasamy from Springer for the continuous support.

On behalf of the EBMT board, we hope this *EBMT Handbook* will be of help in your daily practice.

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Anna Sureda, MD, PhD is nowadays the Head of the Clinical Hematology Department of Institut Català d'Oncologia—Hospitalet, Barcelona and Associate Professor of the University of Barcelona. She had previously been a Senior Consultant in Hospital de la Santa Creu i Sant Pau, Barcelona (from January 1991 to December 2010) and a Senior Consultant focused in lymphomas and hematopoietic cell transplantation (HCT) at Addenbrookes–Cambridge University Hospital, UK (December 2010–December 2012). Anna Sureda has focused her career on clinical investigations into the treatment of Hodgkin's lymphoma, non-Hodgkin's lymphoma, and multiple myeloma patients evaluating novel therapies such as immunotherapy combined with HCT. Anna Sureda was appointed Chairperson of the Lymphoma Working Party (LWP) of the European Group for Blood and Marrow Transplantation (EBMT) from March 2004 to March 2010 and Secretary of the same organization from March 2010 to March 2016. She was elected co-chair of the Lymphoma Committee of the Center for International Blood and Marrow Transplant Research (CIBMTR) and has served the organization in this position from February 2015 to February 2019. She was subsequently appointed as a member of a large non-US Transplant Center in the Advisory Committee of the CIBMTR (from February 2019). Anna Sureda is President of the Spanish Society of Hematopoietic Stem Cell Transplantation and Cellular Therapy (GETH-TC) and, from March 2022, President of the EBMT. Anna Sureda is a regular reviewer of several peer-reviewed journals (*Blood*, *Annals of Oncology*, *Bone Marrow Transplantation*, *The Hematology Journal*, *The European Journal of Hematology* y *Annals of Hematology)* and has co-authored more than 400 manuscripts.



Selim Corbacioglu is Professor and Chair of the Department of Hematology, Oncology and Stem Cell Transplantation at the Children's Hospital in Regensburg, Germany. His major research interest is focused on curative options for hemoglobinopathies and transplant-related systemic endothelial complications. As the PI of several multicenter interventional trials in children, he is the PI for a prospective international trial to evaluate haploidentical HCT in sickle cell disease. He was also one of the lead investigators of the Crisp/Cas9-based gene editing trial in sickle cell disease and thalassemia, where this new approach was successfully applied first-in-man in a patient with thalassemia in his institution. The recipient of the Van Bekkum Award of the EMBT in 2010 for his work on defibrotide prophylaxis in children post-transplant is

author of numerous peer-reviewed articles published in the New England Journal of Medicine, Lancet, Blood, Leukemia and Bone Marrow Transplantation among others. During his term as the scientific council chair and the Chair of the Pediatric Disease Working Party of the EBMT, one of his major contributions was the inauguration of the Hemoglobinopathy Working Party (HGP) of the EBMT and the Dietrich-Niethammer Award for outstanding scientific achievements in pediatric HCT.



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Enric Carreras Degree in Medicine and Surgery (1975), becoming specialist in Internal Medicine and Hematology-Hemotherapy, and Doctor of Medicine (1984) in the University of Barcelona. From 1993 to 2010, he was Director of the HCT Program at the Hospital Clínic in Barcelona and, from 2010 to 2022, Director of the Spanish Bone Marrow Donors Registry at the Josep Carreras Foundation in Barcelona, Spain. During these years, his main clinical/research fields of interest have been focused in early complications and endothelial dysfunction after HCT, HCT in autoimmune diseases, and at-home HCT. He is also founder of the Barcelona Endothelium Team at the Hospital Clinic campus of the Josep Carreras Leukemia Research Institute and the author of more than 400 peer-reviewed international publications mainly focused on HCT. As a member of the EBMT since 1980, he participated actively in the Chronic Leukemia Working Party, ADWP, and Late Complications Working Party. He was Chair of the EBMT/ESH training course in 12 editions and Chair of the EBMT Educational Committee. He is also editor of more than 12 books focused on hematology and HCT, including the four last editions of the EBMT Handbook. In 2018, he was nominated Honorary Member of the EBMT. In 2021, he was awarded with the IACH H.J. Khoury Prize.



Nicolaus Kröger is Professor of Medicine and Medical Director of the Department of Stem Cell Transplantation at the University Medical Center Hamburg-Eppendorf, Germany. He was the President of the EBMT from 2018 to 2022 and Chairman of the German Stem Cell Working Group (DAG-KBT) and Past Chair of the Chronic Malignancies Working Party and of the Scientific Council of EBMT. He is also member of the Scientific Program Committee and the Editorial Board of the European Hematology Association (EHA) and member of the Scientific Committee of the European School for Hematology (ESH). His research interests are on stem cell biology and stem cell transplantation, the detection and treatment of minimal residual disease by adoptive immunotherapy or novel drugs, the

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Abbreviations

AA	Aplastic anemia
Ab	Antibody
ADA	Adenosine deaminase
ADR	Adriamycin
ADV	Adenovirus
Ag	Antigen
aGVHD	Acute graft-versus-host disease
AID	Autoimmune disease
AIHA	Autoimmune hemolytic anemia
AKI	Acute kidney injury
AL	Amyloid light-chain
ALEM	Alemtuzumab
ALG	Antilymphocyte globulin
ALL	Acute lymphoblastic leukemia
Allo-BMT	Allogeneic BMT
Allo-HCT	Allogeneic HCT
AML	Acute myeloid leukemia
ANC	Absolute neutrophil count
APL	Acute promyelocytic leukemia
Ara-C	Cytosine arabinoside
ARDS	Acute or adult respiratory distress syndrome
ASBMT	American Society for Blood and Marrow Transplantation
ATG	Antithymocyte globulin
ATRA	All-trans-retinoic acid
AUC	Area under the curve
Auto-BMT	Autologous BMT
Auto-HCT	Autologous HCT
BAL	Bronchoalveolar lavage
BCNU	1,3-Bis(2-chloroethyl)-1-nitrosourea (carmustine)
BM	Bone marrow
BMDW	Bone Marrow Donors Worldwide
BMF	Bone marrow failure
BMI	Body mass index
BMT	Bone marrow transplantation
BO	Bronchiolitis obliterans
BOOP	Bronchiolitis obliterans organizing pneumonia
BOR	Bortezomib

BOS	Bronchiolitis obliterans syndrome
BSA	Body surface area
BU	Busulfan
BUN	Blood urea nitrogen
BW	Body weight
CAR	Chimeric antigen receptor
CB	Cord blood
CBT	Cord blood transplantation
CBU	Cord blood unit
CC	Complete chimerism
CCI	Charlson Comorbidity Index
CFU	Colony-forming unit
CGD	Chronic granulomatous disease
cGVHD	Chronic graft-versus-host disease
CHF	Congestive heart failure
CI	Comorbidity index
CIBMTR	Center for International Blood and Marrow Transplant Research
CKD	Chronic kidney disease
CLL	Chronic lymphoid/lymphocytic leukemia
CML	Chronic myeloid/myelogenous leukemia
CMML	Chronic myelomonocytic leukemia
CMV	Cytomegalovirus
CMV-IP	CMV-associated interstitial pneumonia
CNI	Calcineurin inhibitor
CNS	Central nervous system
COP	Cryptogenic organizing pneumonia
CR	Complete remission
CR1	First complete remission
CRS	Cytokine release syndrome
CSA	Cyclosporine A
CSF	Cerebrospinal fluid
CT	Computed tomography
CTN	Clinical Trials Network
CVC	Central venous catheter
CVD	Cardiovascular disease
CY	Cyclophosphamide
d	Days
DAH	Diffuse alveolar hemorrhage
DAMP	Damage-associated molecular pattern
DC	Dendritic cell
DEX	Dexamethasone
DFS	Disease-free survival
DIC	Disseminated intravascular coagulation
DLBCL	Diffuse large B-cell lymphoma
DLCL	Diffuse large cell lymphoma
DLCO	Diffusion capacity of the lung for carbon monoxide
DLI	Donor lymphocyte infusion

DLT	Dose-limiting toxicity
DM	Diabetes mellitus
DMSO	Dimethyl sulfoxide
DNA	Deoxyribonucleic acid
DSA	Donor-specific antibody
EBMT	European Society for Blood and Marrow Transplantation
EBNA	Epstein–Barr (virus) nuclear antigen
EBV	Epstein–Barr virus
ECG	Electrocardiogram
ECIL	European Conference on Infections in Leukemia
ECP	Extracorporeal photopheresis
EEG	Electroencephalogram
EFS	Event-free survival
ELISA	Enzyme-linked immunosorbent assay
ELN	European LeukemiaNet
EN	Enteral nutrition
EORTC	European Organisation for Research and Treatment of Cancer
EPO	Erythropoietin
ET	Essential thrombocythemia
EWOG	European Working Group
FA	Fanconi anemia
FACS	Fluorescence-activated cell sorter
FACT	Foundation for the Accreditation of Cellular Therapy
FDA	Food and Drug Administration
FEV1	Forced expiratory volume in 1 s
FFP	Fresh frozen plasma
FFS	Failure-free survival
FISH	Fluorescence in situ hybridization
FL	Follicular lymphoma
FLIPI	Follicular Lymphoma International Prognostic Index
FLU	Fludarabine
FVC	Forced vital capacity
G-CSF	Granulocyte colony-stimulating factor
GF	Graft failure
GFR	Glomerular filtration rate
GI	Gastrointestinal (tract)
GM	Galactomannan
GM-CSF	Granulocyte–macrophage colony stimulated factor
GNB	Gram-negative bacilli
GVH	Graft-versus-host
GVHD	Graft-versus-host disease
GVL	Graft-versus-leukemia
h	Hours
HAART	Highly active antiretroviral therapy
HADS	Hospital Anxiety and Depression Scale
hATG	Horse ATG
HAV	Hepatitis A virus
HBV	Hepatitis B virus

HC	Hemorrhagic cystitis
HCT	Hematopoietic cell transplantation
HCT-CI	HCT-Comorbidity Index
HCV	Hepatitis C virus
HDAC	High-dose Ara-C
HDT	High-dose therapy
HEPA	High-efficiency particulate air
HEV	Hepatitis E virus
HHV	Human herpesvirus
HIB	Haemophilus influenzae type B
HIV	Human immunodeficiency virus
HL	Hodgkin lymphoma
HLA	Human leukocyte antigen
HLH	Hemophagocytic lymphohistiocytosis
HPV	Human papillomavirus
HR	Hazard ratio
HRCT	High-resolution chest tomography
HRT	Hormone replacement therapy
HSC	Hematopoietic stem cell
HSV	Herpes simplex virus
HTLV	Human T-cell lymphotropic virus
HU	Hydroxyurea
HUS	Hemolytic uremic syndrome
HVG	Host-versus-graft
IA	Invasive aspergillosis
IBW	Ideal body weight
ICU	Intensive care unit
IDM	Infectious disease markers
IFI	Invasive fungal infection
IFN	Interferon
Ig	Immunoglobulin
IgG	Immunoglobulin G
IL	Interleukin
IMID	Immunomodulatory drug
IND	Investigational new drug
INR	International normalized ratio
IP	Interstitial pneumonia
IPI	International Prognostic Index
IPS	Idiopathic pneumonia syndrome
IPSS	International Prognostic Scoring System
IRB	Institutional Review Board
IS	Immunosuppressive
IST	Immunosuppressive therapy
ITT	Intent-to-treat
IV	Intravenous
IVIg	Intravenous immunoglobulin
JACIE	Joint Accreditation Committee of ISCT-Europe and EBMT
JCV	JC virus

JMML	Juvenile myelomonocytic leukemia
KIR	Killer immunoglobulin-like receptor
KM	Kaplan–Meier
KPS	Karnofsky Performance Score
L-asp	L-asparaginase
LAF	Laminar air flow
LBL	Lymphoblastic lymphoma
LDH	Lactate dehydrogenase
LENA	Lenalidomide
LFS	Leukemia-free survival
LN	Lymph node
LPS	Lipopolysaccharide
LVEF	Left ventricular ejection fraction
M protein	Monoclonal protein
MA	Myeloablative
MAC	Myeloablative conditioning
MCL	Mantle cell lymphoma
MDS	Myelodysplastic syndrome
MEL	Melphalan
Mesna	Sodium 2-mercaptoethanesulfonate
methylPRD	Methylprednisolone
MF	Myelofibrosis
MFD	Matched family donor
MGUS	Monoclonal gammopathy of undetermined significance
MHC	Major histocompatibility complex
MIC	Minimum inhibitory concentration
min	Minutes
MIPI	Mantle Cell Prognostic Index
miRNA	Micro-RNA
MLC	Mixed leukocyte culture
MM	Multiple myeloma
MMF	Mycophenolate mofetil
MMRD	Mismatched related donor
MMSD	Mismatched sibling donor
MMUD	Mismatched unrelated donor
MoAb	Monoclonal antibody
MODS	Multiple-organ dysfunction syndrome
MOF	Multiorgan failure
MPN	Myeloproliferative neoplasm
MRC	Medical Research Council
MRD	Minimal residual disease
MRI	Magnetic resonance imaging
MRSA	Methicillin-resistant Staphylococcus aureus
MS	Multiple sclerosis
MSC	Mesenchymal stem cell
MSD	Matched sibling donor
MTX	Methotrexate
MUD	Matched unrelated donor

NAT	Nucleic acid amplification test
NC	Nucleated cell
NCI	National Cancer Institute
NGS	Next-generation sequencing
NHL	Non-Hodgkin lymphoma
NIH	National Institutes of Health
NIMA	Non-inherited maternal antigen
NK	Natural killer (cell)
NMA	Non-myeloablative
NMDP	National Marrow Donor Program
NRM	Non-relapse mortality
NSAID	Non-steroidal anti-inflammatory drug
OR	Odds ratio
OS	Overall survival
PAM	Pretransplant assessment of mortality
PB	Peripheral blood
PBSC	Peripheral blood stem cell
PBHCT	Peripheral blood HCT
PCR	Polymerase chain reaction
PERDS	Peri-engraftment respiratory distress syndrome
PET	Positron emission tomography
PFS	Progression-free survival
PFT	Pulmonary function test
Ph	Philadelphia (chromosome)
PHQ-9	Patient Health Questionnaire-9
PICC	Peripherally inserted central venous catheter
PID	Primary immunodeficiency disease
PJP	Pneumocystis jirovecii pneumonia
PK	Pharmacokinetic
PMF	Primary myelofibrosis
PMN	Polymorphonuclear neutrophil
PN	Parenteral nutrition
PNH	Paroxysmal nocturnal hemoglobinuria
PO	Per os
POEMS	Polyneuropathy, organomegaly, endocrinopathy, M protein, skin changes
PR	Partial remission or partial response
PRCA	Pure red cell aplasia
PRD	Prednisone
PRES	Posterior reversible encephalopathy syndrome
PT-CY	Post-HCT cyclophosphamide
PTLD	Post-transplant lymphoproliferative disorder
PUVA	Psoralen–ultraviolet A irradiation
PV	Polycythemia vera
QLQ	Quality of Life Questionnaire
QOL	Quality of life
qRT-PCR	Quantitative reverse transcription polymerase chain reaction
QW	Once weekly

RA	Refractory anemia
RAEB	Refractory anemia with excess blasts
RAEB-T	Refractory anemia with excess blasts in transformation
RARS	Refractory anemia with ringed sideroblasts
RBC	Red blood cell
RCMD	Refractory cytopenia with multilineage dysplasia
RFS	Relapse-free survival
Rh	Rhesus
RI	Relapse incidence
RIC	Reduced-intensity conditioning
R/R	Relapsing/resistant
RR	Relapse rate/Relative risk
RRT	Regimen-related toxicity
RSV	Respiratory syncytial virus
RT-PCR	Real-time polymerase chain reaction
RTx	Radiotherapy
RTX	Rituximab
SAA	Severe aplastic anemia
SARS	Severe acute respiratory syndrome
SC	Subcutaneous
SCD	Sickle cell disease
SCF	Stem-cell factor
SCID	Severe combined immunodeficiency syndrome
SD	Standard deviation
SE	Standard error
SIR	Sirolimus
SLE	Systemic lupus erythematosus
SNP	Single nucleotide polymorphism
SOP	Standard operating procedure
SOS	Sinusoidal obstruction syndrome
SOT	Solid organ transplantation
SPECT	Single-photon emission computed tomography
SR	Standard risk
SS	Sézary syndrome
SSc	Systemic sclerosis
SSOP	Sequence-specific oligonucleotide probe
t-AML	Therapy-related acute myeloid leukemia
t-MDS	Therapy-related myelodysplastic syndrome
TA-GVHD	Transfusion-associated GVHD
TAC	Tacrolimus
TAI	Thoracoabdominal irradiation
TAM	Transplant-associated microangiopathy
TBI	Total body irradiation
TCD	T-cell depletion
TED	Thromboembolic disease
TGF-β	Transforming growth factor beta
THAL	Thalidomide
TKI	Tyrosine kinase inhibitor

TLC	Total lung capacity
TLI	Total lymphoid irradiation
TLR	Toll-like receptor
TLS	Tumor lysis syndrome
TM	Thalassemia major
TMA	Thrombotic microangiopathy
TMP–SMX	Trimethoprim–sulfamethoxazole
TNC	Total nucleated cell
TNF	Tumor necrosis factor
TNF-α	Tumor necrosis factor α
TPN	Total parenteral nutrition
TRALI	Transfusion-related acute lung injury
TREC	T-cell receptor excision circles
Treg	Regulatory T (cell)
TREO	Treosulfan
TRM	Transplant-related mortality
TRT	Transplant-related toxicity
TT	N-Triethylenethiophosphoramide, thioTEPA
TTP	Thrombotic thrombocytopenic purpura
Tx	Therapy/Treatment
UC	Umbilical cord
UCB	Umbilical cord blood
UCBT	Umbilical cord blood transplant
URD	Unrelated donor
UV	Ultraviolet
VCR	Vincristine
VEGF	Vascular endothelial growth factor
VGPR	Very good partial remission
VIN	Vinblastine
VIND	Vindesine
VINO	Vinorelbine
VOD	Veno-occlusive disease
VP	Etoposide
VRE	Vancomycin-resistant *Enterococcus*
vWF	von Willebrand factor
VZIg	Varicella zoster immune globulin
VZV	Varicella zoster virus
WBC	White blood cell
WBMT	Worldwide Network for Blood and Marrow Transplantation
WMDA	World Marrow Donor Association
Wt	Wild-type
X-ALD	X-linked adrenoleukodystrophy
ZAP-70	Zeta-associated protein

Part I

Introduction

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